the same together?

Ans:- Probability of male / female = $\frac{1}{2}$

Probability of all female = Prob(1st female) & Prob(2nd fem) & Prob(3rd female)

$= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$

Probability of all male = $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$

P(3 children same gender) = P(all one gender) =

P(all female) (or) P(all male) = $\frac{1}{8} + \frac{1}{8} = \boxed{\frac{1}{4}}$

(or)

Ex:-9:- What is the probability that 3 children family is 2 girls & one boy?

Ans:- The possible birth orders like (G, G, B) (or) (G, B, G) (or) (B, G, G) and their probabilities like.

$P(G,G,B) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8} = P(G,B,G) = P(B,G,G)$

So Prob(2 girls & one boy) = P(G, G, B) (or) P(G, B, G) (or) P(B, G, G)

$= \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \boxed{\frac{3}{8}}$

~~Ex:-10:~~ [here we can find total probability of all possible occurances as 1.]

Prob (3 girls) = $\frac{1}{8}$, P(2G, 1B) = $\frac{3}{8}$, P(1G, 2B) = $\frac{3}{8}$, P(3G) = $\frac{1}{8}$

Total probabilities = Adding all = $\frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1 = (100\%)$

Ex:-10:- What is the probability that a leap year has 53 Sundays & 52 mondays?

Ans:- Leap year has 52 weeks 2 days., means 52 sun days are already there in those 52 weeks.

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And the remaining 2 days=one of them must sunday and the other is not monday to make leaps year as 53 sundays & 52 mondays.

Total possibilities of last 2 days as:- (i) Sun & Mon. (ii) Mon & Tue (iii) Tue & Wed (iv) Wed & Thu (v) Thu & Fri (vi) Fri & Sat (vii) Sat & Sun

here we have with sun only 2 possibilities (Sun & Mon and Sat & Sun) but we want Sat & Sun (required event) one possibility out of 7 possibilities.

$\Rightarrow P(53 \text{ sundays} \& 52 \text{ mondays}) = \frac{1}{7}$.